

Ali El-Azoiuny (SE2)

Lab-01

A1:

Running the VM:

```
ubuntu@ubuntu-vm: ~  
Microsoft Windows [Version 10.0.19045.6466]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Windows\system32>multipass launch --name ubuntu-vm  
Launched: ubuntu-vm  
  
C:\Windows\system32>  
C:\Windows\system32>multipass shell ubuntu-vm  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-14-generic x86_64)  
  
 * Documentation:  https://docs.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
System information as of Fri Apr 24 15:48:17 EEST 2026  
  
System load:  0.2               Processes:            115  
Usage of /:   56.7% of 3.70GB   Users logged in:     0  
Memory usage: 26%              IPv4 address for eth0: 172.22.108.78  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
2 updates can be applied immediately.  
To see these additional updates run: 'apt list --upgradable'  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
ubuntu@ubuntu-vm:~$ uname -a  
Linux ubuntu-vm 7.0.0-14-generic #14-Ubuntu SMP PREEMPT_DYNAMIC Mon Apr 13 11:09:53 UTC 2026 x86_64 GNU/Linux  
ubuntu@ubuntu-vm:~$
```

Running the container:

```
C:\Windows\system32>docker run -it --name test-container ubuntu /bin/bash  
Unable to find image 'ubuntu:latest' locally  
latest: Pulling from library/ubuntu  
b40150c1c271: Download complete  
Digest: sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc41c7b  
Status: Downloaded newer image for ubuntu:latest  
root@870c16d4ffb1:/# uname -a  
Linux 870c16d4ffb1 5.15.167.4-microsoft-standard-WSL2 #1 SMP Tue Nov 5 00:21:55 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux  
root@870c16d4ffb1:/#
```

A2: Measuring resource overhead and startup time:

VM:

```
ubuntu@ubuntu-vm:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           890Mi       351Mi       199Mi       3.9Mi       479Mi       539Mi
Swap:            0B           0B           0B

ubuntu@ubuntu-vm:~$ ps aux --sort=-%mem | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      1624  0.0  3.5 126456 32688 ?        Ssl   15:45   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root      1524  0.0  3.2  47088 30000 ?        Ss    15:45   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         1  0.1  1.8  25180 16448 ?        Ss    15:45   0:01 /usr/lib/systemd/systemd --switched-root --system --deserialize=44
systemd+  960  0.0  1.6  22744 14840 ?        Ss    15:45   0:00 /usr/lib/systemd/systemd-resolved
root     1535  0.0  1.6  543432 14812 ?       Ssl   15:45   0:00 /usr/libexec/udisks2/udisksd
root      964  0.0  1.4  39336 13556 ?        Ss    15:45   0:00 /usr/lib/systemd/systemd-udevd
root     1666  0.0  1.4  391688 13296 ?       Ssl   15:45   0:00 /usr/sbin/ModemManager
ubuntu   1785  0.0  1.4  22424 12980 ?        Ss    15:45   0:00 /usr/lib/systemd/systemd --user
root     2302  0.0  1.3  18396 12472 ?        Ss    15:59   0:00 sshd-session: ubuntu [priv]

ubuntu@ubuntu-vm:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            179M  932K  178M   1% /run
/dev/sda1        3.8G  2.1G  1.6G  57% /
tmpfs            446M   0  446M   0% /dev/shm
efivarfs         128M   32K  128M   1% /sys/firmware/efi/efivars
tmpfs            446M   0  446M   0% /tmp
none             1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
none             1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda13       909M  106M  810M  12% /boot
/dev/sda15       105M   6.3M   99M   7% /boot/efi
none             1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service
tmpfs            90M   8.0K   90M   1% /run/user/1000
none             1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
none             1.0M   0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
```

```
ubuntu@ubuntu-vm:~$ time multipass start ubuntu -vm
multipass: command not found

real    0m0.122s
user    0m0.069s
sys     0m0.012s
```

Container:

```
root@870c16d4ffb1:/# free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi       882Mi       6.1Gi       3.3Mi       943Mi       6.8Gi
Swap:           2.0Gi           0B       2.0Gi

root@870c16d4ffb1:/# ps aux --sort=-%mem | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root        11  0.0  0.0   7888  3988 pts/0    R+   12:57   0:00 ps aux --sort=-%mem
root         1  0.0  0.0   4588  3972 pts/0    Ss   12:53   0:00 /bin/bash
root        12  0.0  0.0   2708  1096 pts/0    S+   12:57   0:00 head

root@870c16d4ffb1:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay          1007G   11G  946G   2% /
tmpfs            64M   0   64M   0% /dev
tmpfs            3.9G   0   3.9G   0% /sys/fs/cgroup
```

```
PS C:\Windows\system32> Measure-Command { docker start test-container }

Days           : 0
Hours           : 0
Minutes         : 0
Seconds         : 0
Milliseconds    : 533
Ticks           : 5333104
TotalDays       : 6.17257407407407E-06
TotalHours      : 0.000148141777777778
TotalMinutes    : 0.00888850666666667
TotalSeconds    : 0.5333104
TotalMilliseconds : 533.3104
```

Startup Time: container (0.533 S)

VM (0.122 S)

Memory usage: container (used: 882 Mi)

VM (351 MB)

Process count: container (10+ as we used the head command)

VM (3 processes)

B1: launching a free EC2 instance (I used t3.micro as i didn't find the t2 version):

Instance summary for i-06b7c73a091f65ed2 (ubuntu-cloud-vm) info

Updated 3 minutes ago

Instance ID: i-06b7c73a091f65ed2

IPV4 address: --

Hostname type: IP name: ip-172-31-27-165.eu-north-1.compute.internal

Answer private resource DNS name: IPv4 (A)

Auto-assigned IP address: 13.51.161.240 [Public IP]

IAM role: --

IMDSv2: Required

Operator: --

Public IPv4 address: 13.51.161.240 [open address]

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-27-165.eu-north-1.compute.internal

Instance type: t3.micro

VPC ID: vpc-0175acb3b7ef83baa [open address]

Subnet ID: subnet-08fcb26eb79f858c8 [open address]

Instance ARN: arn:aws:ec2:eu-north-1:1871856773320:instance/i-06b7c73a091f65ed2

Private IPv4 addresses: 172.31.27.165

Public DNS: ec2-13-51-161-240.eu-north-1.compute.amazonaws.com [open address]

Elastic IP addresses: --

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. [Learn more]

Auto Scaling Group name: --

Managed: false

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance details info

AMI ID: ami-025d7bea93113b6cc

AMI name: ubuntu/images/hvm-sdd/ubuntu-jammy-22.04-amd64-server-20260218

Monitoring: disabled

Allowed image: --

Platform details: Linux/UNIX

Termination protection: Disabled

Connect with instance:

```
ubuntu@ip-172-31-27-165: ~  
Windows PowerShell  
Copyright (c) Microsoft Corporation. All rights reserved.  
  
Try the new cross-platform PowerShell https://aka.ms/pscore6  
  
PS C:\Users\ZOUNY> cd downloads  
PS C:\Users\ZOUNY\downloads> ssh -i aws-key.pem ubuntu@13.51.161.240  
The authenticity of host '13.51.161.240 (13.51.161.240)' can't be established.  
ED25519 key fingerprint is SHA256:Fict08n+0ZyW+M6zcGrMaScmgIq8enVEYrFDXf6pdS8.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '13.51.161.240' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1046-aws x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Fri Apr 24 13:37:25 UTC 2026  
  
System load:  0.0      Processes:      111  
Usage of /:   22.9% of 7.57GB  Users logged in: 0  
Memory usage: 23%      IPv4 address for ens5: 172.31.27.165  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.
```


B2: Explore Nitro hypervisor (the commands didn't work so explored the nitro architecture {KVM + NVMe = Nitro architecture}):

```
ubuntu@ip-172-31-27-165: $ lscpu | grep Hypervisor
Hypervisor vendor: KVM
ubuntu@ip-172-31-27-165: $ lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0                 7:0      0  27.8M  1 loop /snap/amazon-ssm-agent/12322
loop1                 7:1      0  63.8M  1 loop /snap/core20/2686
loop2                 7:2      0   74M   1 loop /snap/core22/2292
loop3                 7:3      0  91.4M  1 loop /snap/lxd/36918
loop4                 7:4      0  48.1M  1 loop /snap/snapd/25935
nvme0n1              259:0     0    8G   0 disk
├─nvme0n1p1          259:1     0   7.9G   0 part /
├─nvme0n1p14         259:2     0    4M   0 part
└─nvme0n1p15         259:3     0  106M   0 part /boot/efi
ubuntu@ip-172-31-27-165: $
```

B3: Htop running:

```
0% Tasks: 29, 31 thr: 1 running
1% Load average: 0.24 0.10 0.03
Mem[|||||] 195M/914M Uptime: 00:20:16
Swap[ ] 0K/0K

  PID USER   PRI  NI  VIRT   RES   SHR  CPU MEM%   TIME+  Command
1735 ubuntu  20    0  8992  4860  3512  0.0  0.5  0:00.04 htop
1 root    20    0  163M  12976  6420  0.0  1.4  0:01.41 /sbin/init
124 root   19    0  47748 14312 13232  0.0  1.5  0:00.20 /lib/systemd/systemd-journald
166 root   RT    0  282M  27624  8072  0.0  3.0  0:00.11 /sbin/multipathd -d -s
170 root   20    0  11364  6240  4496  0.0  0.7  0:00.17 /lib/systemd/systemd-udev
171 root   20    0  282M  27624  8072  0.0  3.0  0:00.00 /sbin/multipathd -d -s
172 root   RT    0  282M  27624  8072  0.0  3.0  0:00.00 /sbin/multipathd -d -s
173 root   RT    0  282M  27624  8072  0.0  3.0  0:00.00 /sbin/multipathd -d -s
174 root   RT    0  282M  27624  8072  0.0  3.0  0:00.00 /sbin/multipathd -d -s
175 root   RT    0  282M  27624  8072  0.0  3.0  0:00.06 /sbin/multipathd -d -s
176 root   RT    0  282M  27624  8072  0.0  3.0  0:00.00 /sbin/multipathd -d -s
347 systemd-n 20    0  16260  6640  7580  0.0  0.9  0:00.04 /lib/systemd/systemd-networkd
349 systemd-r 20    0  26336 14288  8188  0.0  1.5  0:00.09 /lib/systemd/systemd-resolved
400 root    20    0  2816  2136  1988  0.0  0.2  0:00.00 /usr/sbin/acpid
404 root    20    0  7288  2932  2672  0.0  0.3  0:00.00 /usr/sbin/cron -f -P
405 messagebu 20    0  8596  4960  4244  0.0  0.5  0:00.09 @dbus-daemon --system --address=systemd: --nofork --nopidfile --system
413 root    20    0  81704  4156  3792  0.0  0.4  0:00.03 /usr/sbin/irqbalance --foreground
414 root    20    0  33096 18516 18500  0.0  2.1  0:00.09 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
416 syslog   20    0  217M  8036  4620  0.0  0.6  0:00.02 /usr/sbin/rsyslogd -n -iNONE
418 root    20    0  1366M 38256 24576  0.0  4.1  0:00.66 /usr/lib/snapd/snapd
419 root    20    0  81704  4156  3792  0.0  0.4  0:00.00 /usr/sbin/irqbalance --foreground
424 root    20    0  11336  7880  6876  0.0  0.8  0:00.10 /lib/systemd/systemd-logind
432 _chrony   20    0  10908  3772  3064  0.0  0.4  0:00.05 /usr/sbin/chronyd -F 1
435 _chrony   20    0  10580  3184  1612  0.0  0.2  0:00.00 /usr/sbin/chronyd -F 1
442 syslog   20    0  217M  8036  4620  0.0  0.6  0:00.00 /usr/sbin/rsyslogd -n -iNONE
443 syslog   20    0  217M  8036  4620  0.0  0.6  0:00.00 /usr/sbin/rsyslogd -n -iNONE
444 syslog   20    0  217M  8036  4620  0.0  0.6  0:00.00 /usr/sbin/rsyslogd -n -iNONE
468 root    20    0  1366M 38256 24576  0.0  4.1  0:00.02 /usr/lib/snapd/snapd
469 root    20    0  1366M 38256 24576  0.0  4.1  0:00.00 /usr/lib/snapd/snapd
470 root    20    0  1366M 38256 24576  0.0  4.1  0:00.09 /usr/lib/snapd/snapd
471 root    20    0  1366M 38256 24576  0.0  4.1  0:00.00 /usr/lib/snapd/snapd
486 root    20    0  6220  2400  2252  0.0  0.3  0:00.00 /sbin/agetty -o -p -- \u --keep-baud 115200,57600,38400,9600 ttyS0 vt
497 root    20    0  6176  2344  2208  0.0  0.3  0:00.00 /sbin/agetty -o -p -- \u --noclear tty1 linux
517 root    20    0  107M  21900 13644  0.0  2.3  0:00.09 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-sh
584 root    20    0  229M  6600  7232  0.0  0.9  0:00.02 /usr/libexec/polkitd --no-debug
589 root    20    0  1366M 38256 24576  0.0  4.1  0:00.10 /usr/lib/snapd/snapd
594 root    20    0  229M  6600  7232  0.0  0.9  0:00.00 /usr/libexec/polkitd --no-debug
596 root    20    0  107M  21900 13644  0.0  2.3  0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-sh
597 root    20    0  229M  6600  7232  0.0  0.9  0:00.00 /usr/libexec/polkitd --no-debug
601 root    20    0  1366M 38256 24576  0.0  4.1  0:00.00 /usr/lib/snapd/snapd
625 root    20    0  1366M 38256 24576  0.0  4.1  0:00.10 /usr/lib/snapd/snapd
632 root    20    0  15440  3404  7772  0.0  1.0  0:00.01 sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ec2-insta
655 root    20    0  1366M 38256 24576  0.0  4.1  0:00.14 /usr/lib/snapd/snapd
656 root    20    0  1366M 38256 24576  0.0  4.1  0:00.05 /usr/lib/snapd/snapd
1089 root    20    0  1715M 28192 11644  0.0  2.2  0:00.72 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1103 root    20    0  1715M 28192 11644  0.0  2.2  0:00.00 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1104 root    20    0  1715M 28192 11644  0.0  2.2  0:00.00 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1105 root    20    0  1715M 28192 11644  0.0  2.2  0:00.00 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1106 root    20    0  1715M 28192 11644  0.0  2.2  0:00.00 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1112 root    20    0  1715M 28192 11644  0.0  2.2  0:00.04 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1113 root    20    0  1715M 28192 11644  0.0  2.2  0:00.12 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1115 root    20    0  1715M 28192 11644  0.0  2.2  0:00.01 /snap/amazon-ssm-agent/12322/amazon-ssm-agent
1159 root    20    0  17056 11084  8912  0.0  1.2  0:00.01 sshd: ubuntu [priv]
1162 ubuntu  20    0  17048  8804  8176  0.0  1.0  0:00.05 /lib/systemd/systemd --user
1163 ubuntu  20    0  166M  6500  1672  0.0  0.7  0:00.00 (sd-pam)
1242 ubuntu  20    0  17192  8100  5660  0.0  0.9  0:00.02 sshd: ubuntu@pts/0
1245 ubuntu  20    0  9100  5340  3588  0.0  0.6  0:00.04 -bash
1631 root    20    0  289M  28572 17532  0.0  2.2  0:00.02 /usr/libexec/packagekitd
1632 root    20    0  289M  28572 17532  0.0  2.2  0:00.00 /usr/libexec/packagekitd
1633 root    20    0  289M  28572 17532  0.0  2.2  0:00.00 /usr/libexec/packagekitd

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Vice F9Kill F10Quit
```

C1: py app running:



C2: tests:

```
PS C:\Apache24\bin> .\ab -n 100 -c 10 http://localhost:5000/
This is ApacheBench, Version 2.3 <$Revision: 1923142 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking localhost (be patient).....done

Server Software:        Werkzeug/3.1.8
Server Hostname:        localhost
Server Port:            5000

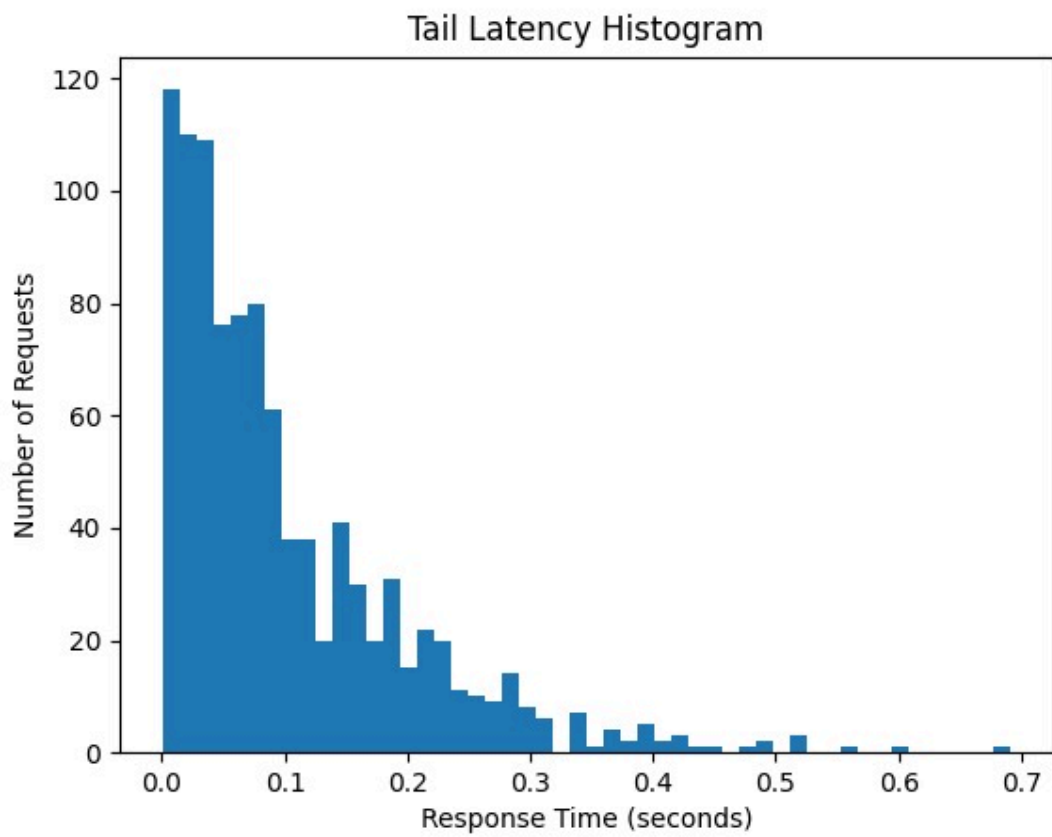
Document Path:          /
Document Length:        27 bytes

Concurrency Level:      10
Time taken for tests:    3.507 seconds
Complete requests:      100
Failed requests:         0
Total transferred:      20000 bytes
HTML transferred:       2700 bytes
Requests per second:    28.52 [#/sec] (mean)
Time per request:       350.650 [ms] (mean)
Time per request:       35.065 [ms] (mean, across all concurrent requests)
Transfer rate:          5.57 [Kbytes/sec] received

Connection Times (ms)
              min    mean[+/-sd] median   max
Connect:        0      0   0.5      0      1
Processing:    12    123  91.1    108    499
Waiting:        2    104  91.1     86    474
Total:         13    123  91.1    109    500

Percentage of the requests served within a certain time (ms)
 50%    109
 66%    140
 75%    158
 80%    187
 90%    250
 95%    327
 98%    375
 99%    500
100%    500 (longest request)
```

Latency histogram:



Discussion Prompts:

1) Why does AWS split Nitro into hardware components?

- It's split mainly for 3 reasons:

a) Performance: Offloading heavy tasks like networking, storage, and management to custom hardware cards frees up almost 100% of the main server CPU and memory for your actual workloads.

b) Security: Physically isolating the management layer reduces the attack surface and locks down the system so tightly that even AWS administrators cannot access your data.

c) Modularity: Treating these components like building blocks allows AWS to mix and match hardware to release new, specialized EC2 instance types much faster.

2) In which scenarios would you use VMs over containers?

-for 3 reasons:

a) Strict Security: When you need hardware-level isolation between applications.

b) Different OS: When the app requires a different operating system than the host (e.g., Windows on Linux).

c) Legacy Software: When older apps require a full OS environment or custom kernel features.

3) How does tail latency change with the number of parallel calls?

-Tail latency increases. More parallel calls mean a higher probability of waiting on at least one unusually slow response, which delays the entire system.